1. Write programs in Python to display following sequences for n terms (for loop):	2. Write programs in Python to find and display the sum of the following series for n terms (for loop):
(a) **1, 2, 3, ... N Terms**	(a) **1+ 2+ 3+ ... N Terms**
(b) **2, 4, 6, ... N Terms**	(b) **2+ 4+ 6+ ... N Terms**
(c) **1, 5, 9, ... N Terms**	(c) **1+ 5+ 9+ ... N Terms**
(d) **X, X^2, X^3, ... N Terms**	(d) **X+ X^2+ X^3+ ... N Terms**
(e) **X, X^2/2!, X^3/3!, ... N Terms**	(e) **X+ X^2/2! + X^3/3!+ ... N Terms**
(f) **X, -X^2/2!, X^3/3!, ... N Terms**	(f) **X-X^2/2!+ X^3/3! - ... N Terms**
(g) **1, (1+2), (1+2+3), ... N Terms**	(g) **1+(1+2)+(1+2+3)+ ... N Terms**
(h) **2, (2+4), (2+4+6), ... N Terms**	(h) **2+ (2+4)+ (2+4+6)+ ... N Terms**

3. Write a program in Python to check if a number (integer) entered by the user is PRIME or not.
4. Write a program in Python to check if a number (integer) entered by the user is COMPOSITE or not.
5. Write a program in Python to print first N PRIME numbers, where the value of N to be entered by the user.
6. Write a program in Python to print PRIME numbers between 1 to N, where the value of N to be entered by the user.
7. Write a program in Python to check if an integer N entered by the user is an **ARMSTRONG** number of not.

   Note: The **ARMSTRONG** number is a number, whose sum of individual digits raised to the power of the digit count is equal to the original number itself . Examples: **0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 153, 370, 371, 407, 1634** etc. are armstrong numbers
   As 1^1=1, 9^1=9, $1^3+5^3+3^3$=153, $1^4+6^4+3^4+4^4$=1634
   There is no 2 digit ARMSTRONG number

8. Write a program to print all the **ARMSTRONG** numbers between 0 to N, where the value of N to be entered by the user.
9. Write a program in Python to reverse digits of an integer N and display the digits in the reversed order in the same line with each digit ending with # .

```
Enter N:345
5 # 4 # 3 #
```

10. Write a program in Python to find a new number, which carries reversed digits of an integer N.

```
Enter N:345
M (Reversed content of N): 543
```

11. Write a program in Python to check if a new number N entered by the user is Palindrome or not. Palindrome is a number, which remains the same when its digits are reversed.

```
Enter N:345                          Enter N: 12321
345 is not a palindrome              12321 is a Palindrome
More numbers to check (Y/N)? Y       More numbers to check (Y/N)? N
```

12. Write a program in Python to add those digits of an integer N, which are on odd positions.

```
Enter N:1834                         Enter N: 24192
Sum of odd positions:4               Sum of odd positions:5
More numbers (Y/N):Y                 More numbers (Y/N):N
```

13. Write a program in Python to find all the factors of a given number (integer).

```
Enter N:18                           Enter N: 24
1,2,3,6,9,18                         1,2,3,4,6,8,12,24
Factors required for More(Y/N)? Y    Factors required for More(Y/N)? N
```

14. Write a program in Python to find all the HCF of two given numbers (integers).

```
Enter N:12                          Enter N: 24
Enter M:18                          Enter M: 16
HCF    :6                           HCF    : 8
HCF required for More(Y/N)? Y       HCF required for More(Y/N)? N
```

15. Write a program in Python to find the LCM of two given numbers (integers).

```
Enter N:12                          Enter N: 24
Enter M:18                          Enter M: 16
LCM    :36                          HCF    : 48
LCM required for More(Y/N)? Y       LCM required for More(Y/N)? N
```

16. Write a program in Python to perform operations (Addition, Subtraction, Multiplication, Division) on several sets of two integers.

```
Enter A:12                          Enter A:120
Enter B:18                          Enter B:30
Operation +,-,*,/ :+                Operation +,-,*,/ :*
Result: 30                          Result: 3600
More Operations (Y/N)? Y            More Operations (Y/N)? N
```

17. Write a program in Python to find the area of various 2D objects as per the user's choice.

```
Object Shape [S:Square R:Rectangle C:Circle Q:Quit]:S
Enter Side:12
Area of Square:144
Object Shape [S:Square R:Rectangle C:Circle Q:Quit]:R
Enter Length:12
Enter Breadth:8
Area of Rectangle:96
Object Shape [S:Square R:Rectangle C:Circle Q:Quit]:C
Enter Radius:5
Area of Circle:78.54
Object Shape [S:Square R:Rectangle C:Circle Q:Quit]:Q
Thanks
```

IMPORTANT: REFER TO MORE QUESTIONS GIVEN IN ASSIGNMENT BOOKLET

General Instructions:

i. Type the solutions of all the problems using **Python** Language on **Python IDLE** or **colab.research.google.com**

ii. Type in the following on top of each of your program as comment lines

```
# ------------------------------------------------------------#
# List-Program No     : L2-P1
# Developed By        : Jhilmil Roy
# Date                : 16-Jul-2024
# ------------------------------------------------------------#
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample output at the bottom of the program as comment lines.

iv. **"Turn in"** the softcopy of the programs as a single document in response to this assignment submission with each program on separate pages. Once verified by the teacher, take out a hardcopy of **each program on separate pages** from the printer and get them signed by the respective computer teacher. Add all the printouts in the practical file and get the Index entry also signed.

v. Recommended Font, "**Courier New**", Style "**Bold**" Size "**10**" for all programs with single line spacing. Default indentation **2 Character** only.

```
# ------------------------------------------------------------#
# List-Program No    : L2-P1
# Developed By       : PUJA DEY
# Date               : 10-Jul-2024
# ------------------------------------------------------------#

print("1, 2, 3, ... N Terms")
N = int(input("Enter N: "))
for i in range(1,N+1):
 print(i, end="  ")
print("\n")

print("2, 4, 6, ... N Terms")
N = int(input("Enter N: "))
for i in range(2, 2*N+2, 2):
 print(i, end="  ")
print("\n")

print("1, 5, 9, ... N Terms")
N = int(input("Enter N: "))
for i in range(1, 4*N+1, 4):
 print(i, end="  ")
print("\n")

print("X, X^2, X^3, ... N Terms")
X = int(input("Enter X: "))
N = int(input("Enter N: "))
for i in range(1,N+1):
 print(X**i, end="  ")
print("\n")

print("X, X^2/2!, X^3/3!, ... N Terms")
X = int(input("Enter X: "))
N = int(input("Enter N: "))
for i in range(1,N+1):
 f=1
 for z in range(1,i+1):
   f*=z
 print((X**i)/f, end="  ")
print("\n")
```

```
print("X, -X^2/2!, X^3/3!, ... N Terms")
X = int(input("Enter X: "))
N = int(input("Enter N: "))
for i in range(1,N+1):
 sign = ((i%2)*2)-1
 f=1
 for z in range(1,i+1):
   f*=z
 print(sign*(X**i)/f, end="  ")
print("\n")

print("1, (1+2), (1+2+3), ... N Terms")
N = int(input("Enter N: "))
for i in range(1,N+1):
 s = 0
 for k in range(1,i+1):
   s+=k
 print(s, end="  ")
print("\n")

print("2, (2+4), (2+4+6), ... N Terms")
N = int(input("Enter N: "))
for i in range(1, N+1):
 s = 0
 for k in range(1,i+1):
   s+=2*k
 print(s, end="  ")
print("\n")

1, 2, 3, ... N Terms
Enter N: 10
1  2  3  4  5  6  7  8  9  10

2, 4, 6, ... N Terms
Enter N: 10
2  4  6  8  10  12  14  16  18  20

1, 5, 9, ... N Terms
Enter N: 10
```

```
1  5  9  13  17  21  25  29  33  37

X, X^2, X^3, ... N Terms
Enter X: 10
Enter N: 10
10  100  1000  10000  100000  1000000  10000000  100000000  1000000000  10000000000

X, X^2/2!, X^3/3!, ... N Terms
Enter X: 10
Enter N: 10
10.0  50.0  166.66666666666666  416.6666666666667  833.3333333333334
1388.888888888889  1984.126984126984  2480.15873015873  2755.731922398589
2755.731922398589

X, -X^2/2!, X^3/3!, ... N Terms
Enter X: 10
Enter N: 10
10.0  -50.0  166.66666666666666  -416.6666666666667  833.3333333333334
-1388.888888888889  1984.126984126984  -2480.15873015873  2755.731922398589
-2755.731922398589

1, (1+2), (1+2+3), ... N Terms
Enter N: 10
1  3  6  10  15  21  28  36  45  55

2, (2+4), (2+4+6), ... N Terms
Enter N: 10
2  6  12  20  30  42  56  72  90  110
```

```
# ------------------------------------------------------------------#
# List-Program No   : L2-P2
# Developed By    : PUJA DEY
# Date            : 10-July-2024
# ------------------------------------------------------------------#

s=0
print("1+ 2+ 3+ ... N Terms")
N = int(input("Enter N: "))
for i in range(1,N+1):
 s+=i
print(s, "\n")

s=0
print("2+ 4+ 6+ ... N Terms")
N = int(input("Enter N: "))
for i in range(2, 2*N+2, 2):
 s+=i
print(s,"\n")

s=0
print(" 1+ 5+ 9+ ... N Terms")
N = int(input("Enter N: "))
for i in range(1, 4*N+1, 4):
 s+=i
print(s,"\n")

s=0
print("X+ X2+ X3+ ... N Terms")
X = int(input("Enter X: "))
N = int(input("Enter N: "))
for i in range(1,N+1):
 s+=X**i
print(s,"\n")

s=0
print("X+ X2/2! +  X3/3!+ ... N Terms")
X = int(input("Enter X: "))
N = int(input("Enter N: "))
for i in range(1,N+1):
```

```
 f=1
 for z in range(1,i+1):
   f*=z
 s+=(X**i)/f
print(s,"\n")

s=0
print("X-X2/2!+ X3/3! - ... N Terms")
X = int(input("Enter X: "))
N = int(input("Enter N: "))
for i in range(1,N+1):
 sign = ((i%2)*2)-1
 f=1
 for z in range(1,i+1):
   f*=z
 s+= sign*(X**i)/f
print(s,"\n")

s=0
print("1+(1+2)+(1+2+3)+ ... N Terms")
N = int(input("Enter N: "))
for i in range(1,N+1):
 l = 0
 for k in range(1,i+1):
   l+=k
 s+=l
print(s,"\n")

s=0
print("2+ (2+4)+ (2+4+6)+ ... N Terms")
N = int(input("Enter N: "))
for i in range(1, N+1):
 l = 0
 for k in range(1,i+1):
   l+=2*k
 s+=l
print(s,"\n")

1+ 2+ 3+ ... N Terms
Enter N: 7
28
```

```
2+ 4+ 6+ ... N Terms
Enter N: 7
56

 1+ 5+ 9+ ... N Terms
Enter N: 6
66

X+ X2+ X3+ ... N Terms
Enter X: 4
Enter N: 5
1364

X+ X2/2! +  X3/3!+ ... N Terms
Enter X: 3
Enter N: 6
18.412499999999998

X-X2/2!+ X3/3! - ... N Terms
Enter X: 7
Enter N: 3
39.666666666666664

1+(1+2)+(1+2+3)+ ... N Terms
Enter N: 5
35

2+ (2+4)+ (2+4+6)+ ... N Terms
Enter N: 6
112
```

```
# ------------------------------------------------------------------#
# List-Program No   : L2-P3
# Developed By     : PUJA DEY
# Date             : 10-July-2024
# ------------------------------------------------------------------#
n= int(input("Enter Number: "))
for i in range(2,n//2+1):
  if n%i==0:
    print(n,"is not a prime")
    break
else:
  print(n, "is a prime")
```

```
Enter Number: 5
5 is a prime
```

```
# ------------------------------------------------------------------#
# List-Program No   : L2-P4
# Developed By    : PUJA DEY
# Date            : 10-July-2024
# ------------------------------------------------------------------#

n= int(input("Enter Number: "))

for i in range(2,n//2+1):
 if n%i==0:
   print(n,"is composite")
   break
else:
 print(n, "is a NOT composite")
```

```
Enter Number: 90
90 is composite
```

```
# ------------------------------------------------------------------#
# List-Program No   : L2-P5
# Developed By    : PUJA DEY
# Date            : 10-July-2024
# ------------------------------------------------------------------#

N=int(input("First N prime number will be printed (Enter N): "))
n=2
p=0
while True:
 for i in range(2,n//2+1):
   if n%i==0:
     break
 else:
   print(n, end="  ")
   p+=1

 n+=1

 if p == N:
   break
```

```
First N prime number will be printed (Enter N): 7
2  3  5  7  11  13  17
```

```
# ------------------------------------------------------------------#
# List-Program No    : L2-P6
# Developed By     : PUJA DEY
# Date             : 10-July-2024
# ------------------------------------------------------------------#

N = int(input("Prime numbers will be printed from 1 to N: "))
for n in range(2,N+1):
 for i in range(2,n//2+1):
   if n%i==0:
     break
 else:
   print(n, end="  ")

Prime numbers will be printed from 1 to N: 69
2  3  5  7  11  13  17  19  23  29  31  37  41  43  47  53  59  61  67
```

```
# ------------------------------------------------------------#
# List-Program No   : L2-P7
# Developed By    : PUJA DEY
# Date            : 10-July-2024
# ------------------------------------------------------------#

c = 0
n = int(input("Enter a number to check if it is an armstrong number: "))
i=n
while i >=1:
 i = i/10
 c+=1

s=0
p=n
for k in range(c):
 s+=(p%10)**c
 p=p//10

if s==n:
 print(n,"is an armstrong number")
else:
 print(n, "is not an armstrong number")

Enter a number to check if it is an armstrong number: 1633
1633 is not an armstrong number
```

```
# ------------------------------------------------------------#
# List-Program No   : L2-P8
# Developed By    : PUJA DEY
# Date            : 10-July-2024
# ------------------------------------------------------------#

k=int(input("Enter number (All numbers from 0 to N which are armstrong numbers will be printed):"))
for n in range(0,k+1):
 c = 0
 i=n
 while i >=1:
   i = i/10
   c+=1

 s=0
 p=n
 for w in range(c):
   s+=(p%10)**c
   p=p//10

 if s==n:
   print(n, end="   ")
```

```
Enter number (All numbers from 1 to N which are armstrong numbers will be printed):2000
0   1   2   3   4   5   6   7   8   9   153   370   371   407   1634
```

```
# ------------------------------------------------------------#
# List-Program No   : L2-P9
# Developed By    : PUJA DEY
# Date            : 10-July-2024
# ------------------------------------------------------------#

c = 0
n = int(input("Input a number (Number will be reversed and printed): "))
i=n
while i >=1:
 i = i/10
 c+=1

for k in range(1,c+1):
 print(n%10, end=" # ")
 n=n//10


Input a number (Number will be reversed and printed): 457239
9 # 3 # 2 # 7 # 5 # 4 #
```

```
# ------------------------------------------------------------#
# List-Program No   : L2-P10
# Developed By    : PUJA DEY
# Date            : 10-July-2024
# ------------------------------------------------------------#

c = 0
n = int(input("Input a number (Number will be reversed and printed): "))
i=n
while i >=1:
 i = i/10
 c+=1

s=0
for k in range(1,c+1):
 s+=(n%10)*(10**(c-k))
 n=n//10

print(s)

Input a number (Number will be reversed and printed): 567
765
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P11
# Developed By       : PUJA DEY
# Date               : 10-Jul-2024
# ------------------------------------------------------------#

while True:
 A=input("Enter Num: ")
 B=str(A)
 if B[::]==B[::-1]:
    print("Number is a Palindrome")
 else:
    print("Number is  not a Palindrome")
 ch=input("Required for more numbers (Y/N)?: ")
 if ch=="N":
    break

Enter Num: 345
Number is  not a Palindrome
Required for more numbers (Y/N)?: Y
Enter Num: 12321
Number is a Palindrome
Required for more numbers (Y/N)?: N
```

```
# ------------------------------------------------------------------#
# List-Program No    : L2-P12
# Developed By       : PUJA DEY
# Date               : 10-Jul-2024
# ------------------------------------------------------------------#

ch="Y"
while ch=="Y":
 S=0
 A=input("Enter Num: ")
 B=str(A)
 for i in range(len(B)):
  if i%2 ==0:
   S+=int(B[i])
 print("Sum of odd positions: ", S)
 ch=input("Required for more numbers (Y/N)?: ")

Enter Num: 1834
Sum of odd positions:  4
Required for more numbers (Y/N)?: Y
Enter Num: 24192
Sum of odd positions:  5
Required for more numbers (Y/N)?: N
```

```
# ------------------------------------------------------------#
# List-Program No   : L2-P13
# Developed By    : PUJA DEY
# Date            : 10-July-2024
# ------------------------------------------------------------#

ch="Y"
while ch=="Y" or ch=="y":
  N=int(input("Enter a number: "))
  print("Factors are: ", end="")
  for i in range (1, N+1):
    if N%i==0:
      print(i, end=",")
  print()
  ch=input("Factors required for more numbers (Y/N)?")
Enter a number: 18
Factors are: 1,2,3,6,9,18,
Factors required for more numbers (Y/N)?y
Enter a number: 24
Factors are: 1,2,3,4,6,8,12,24,
Factors required for more numbers (Y/N)?n
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P14
# Developed By       : PUJA DEY
# Date               : 10-Jul-2024
# ------------------------------------------------------------#

while True:
 A=int(input("Enter Num1: "))
 B=int(input("Enter Num2: "))
 HCF=1
 MIN=min(A,B)
 for I in range(1,MIN+1):
  if A%I==0 and B%I==0:
    HCF=I
 print("HCF of",A,"and",B,"is",HCF)
 ch=input("Required for more numbers (Y/N)?: ")
 if ch=="N":
    break
Enter Num1: 12
Enter Num2: 18
HCF of 12 and 18 is 6
Required for more numbers (Y/N)?: Y
Enter Num1: 24
Enter Num2: 16
HCF of 24 and 16 is 8
Required for more numbers (Y/N)?: N
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P15
# Developed By       : PUJA DEY
# Date               : 10-Jul-2024
# ------------------------------------------------------------#
while True:
 A=int(input("Enter Num1: "))
 B=int(input("Enter Num2: "))
 HCF=1
 MIN=min(A,B)
 for I in range(1,MIN+1):
  if A%I==0 and B%I==0:
    HCF=I
 print("LCM of",A,"and",B,"is",A*B//HCF)
 ch=input("Required for more numbers (Y/N)?: ")
 if ch=="N":
    break
Enter Num1: 12
Enter Num2: 18
LCM of 12 and 18 is 36
Required for more numbers (Y/N)?: Y
Enter Num1: 24
Enter Num2: 16
LCM of 24 and 16 is 48
Required for more numbers (Y/N)?: N
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P16
# Developed By       : PUJA DEY
# Date               : 10-Jul-2024
# ------------------------------------------------------------#

while True:
  A=int(input("Enter A: "))
  B=int(input("Enter B: "))
  Op=input("Operation [+,-,*,/]: ")
  if Op=="+":
    print("Result: ", A+B)
  elif Op=="-":
    print("Result: ", A-B)
  elif Op=="*":
    print("Result: ", A*B)
  elif Op=="/":
    print("Result: ", A/B)
  else:
    print("Invalid")
  ch=input("Required for more operations (Y/N)?: ")
  if ch=="N":
    break
Enter A: 12
Enter B: 18
Operation [+,-,*,/]: +
Result:  30
Required for more operations (Y/N)?: Y
Enter A: 12
Enter B: 300
Operation [+,-,*,/]: *
Result:  3600
Required for more operations (Y/N)?: N
```

```
# ------------------------------------------------------------#
# List-Program No    : L2-P17
# Developed By       : PUJA DEY
# Date               : 10-Jul-2024
# ------------------------------------------------------------#

while True:
  Ch=input("Object Shape [S:Square R:Rectangle C:Circle Q:Quit]: ")
  if Ch=="S":
      side=int(input("Enter Side: "))
      Area=side**2
      print("Area of Square: ", Area)
  elif Ch=="R":
      L=int(input("Enter Length: "))
      B=int(input("Enter Breadth: "))
      AreaR=L*B
      print("Area of Rectangle: ", AreaR)
  elif Ch=="C":
      r=int(input("Enter Radius: "))
      AreaC=3.14*(r**2)
      print("Area of Circle: ", AreaC)
  else:
      print("Thanks")
      break
Object Shape [S:Square R:Rectangle C:Circle Q:Quit]: S
Enter Side: 12
Area of Square:  144
Object Shape [S:Square R:Rectangle C:Circle Q:Quit]: R
Enter Length: 12
Enter Breadth: 8
Area of Rectangle:  96
Object Shape [S:Square R:Rectangle C:Circle Q:Quit]: C
Enter Radius: 5
Area of Circle:  78.5
Object Shape [S:Square R:Rectangle C:Circle Q:Quit]: Q
Thanks
```